STOWAWAY / ENGINEERING DECISION REPORT

Multi-source architecture and EPUB foundation

Decision report and bounded implementation handoff

Prepared for Owen Chapman / 10 October 2026

READY FOR MULTI-SOURCE FOUNDATION

Keep the accepted reader. Separate acquisition pipelines. Start with a dependency-free capability and policy boundary.

REVIEWED REPOSITORY

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main d43cf0963497e40836cac40f0c2ed2df43a0262d

Read-only review of implementation, accepted project evidence and current official sources. No branch, commit, PR or implementation was created.

Includes the complete coupling map, TypeScript contracts, minimal SQL migration, EPUB acceptance limits, source gating, dependency decisions and a reusable M0 implementation prompt.

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How to read this report

The numbered sections follow the requested decision format. Sections 15-17 define the only proposed first implementation slice. Code and SQL are illustrative proposals, not repository changes.

Evidence labels [R1-R15] refer to the pinned repository; [E1-E24] refer to official external sources. The evidence register follows the handoff. Recommendations and proposed limits are identified separately from verified implementation facts.

1. EXECUTIVE VERDICT

READY FOR MULTI-SOURCE FOUNDATION

Stowaway can accept a second origin without replacing its reader or rewriting its accepted SpaceBattles product. Choose **distinct remote-source and local-file ingestion pipelines that converge on a small normalized manifest, the existing canonical chapter representation, and validated local assets**.

Start with **M0: a capability and policy boundary around the existing services**. It adds no source, database migration or dependency. It makes source eligibility explicit, routes screen actions through guarded dispatch, and introduces an explicit internal/public build audience. Existing SpaceBattles acquisition remains available in the intended internal build. EPUB parsing, persistence migration and reader changes belong to subsequent, separately verified slices.

The strongest evidence is in the implementation. Chapter and Locator already use source-neutral string identifiers and canonical HTML/hash/block contracts. Most reader, availability, Read state and continuation logic can remain shared. The important leaks are the required thread/post fields, pending chapter-ID synthesis, ownership keys parsed from ":post:", SpaceBattles-specific repository writers, image acquisition, and screen action checks. These are bounded changes before a second source, rather than evidence of a failed core architecture. [R2-R10]

Decisions to carry forward

- Preserve all existing SpaceBattles IDs, canonical HTML, hashes, sanitizer versions, block IDs, locators, Read flags, availability and ownership/quiescence behavior.
- Treat acquisition method and declared content provenance as separate facts. An EPUB mentioning AO3 remains a local file import; it never gains live AO3 capabilities.
- Give arbitrary file imports persistent UUID identities. Use a package SHA-256 as a byte-version and duplicate hint, never as the sole work identity.
- Keep historical SQL names temporarily behind a documented mapper. For EPUB persistence, add explicit manifest chapter IDs and small package/entry provenance sidecars; avoid parent-table rebuilds.
- Retain an app-owned original EPUB. Restoration is deliberate local processing, with truthful labels and no network fallback.
- Target a constrained, reflowable EPUB2/3 importer. The initial proof does not claim full EPUB reading-system conformance or working internal hyperlinks.
- Complete EPUB proof, then backup/restore, then source-specific official-download handling, then only approved direct adapters.
- Keep public distribution blocked until the actual public composition and artifact have been reviewed. M0's service guard is useful enforcement, but it is not proof that disallowed source code is absent from a binary.

Evidence and confidence

Repository fact: Connected GitHub main matched the handoff SHA. The current "Stowaway checks" run 38022524764 for that SHA completed successfully. Repository documents record approximately 829 tests across 30 suites and Owen's standalone iPhone acceptance. This review did not rerun the application suite or observe the phone. Historical unexercised cases remain unexercised. [R1, R11-R15]

Review work: The team read all thirteen requested documents, inspected all 64 production source files and the current tests/fixtures, and traced the source, storage, ownership, image and reader paths. An isolated in-memory SQLite probe exercised the proposed leaf-table migration against the exact six current migration blocks and synthetic state. It demonstrated SQL feasibility, preservation and rollback on host SQLite 3.53.1; it did not validate an Expo native migration.

Official external fact: Research used official specifications, documentation, package repositories and source policies as accessed on 10 October 2026. EPUB 3.3 is a W3C Recommendation; EPUB 3.4 is currently a Candidate Recommendation Draft. [E1, E5]

Engineering inference: The canonical reading model is suitable for a second origin. The principal new technical uncertainty is safe, bounded archive/XML work on the actual Expo57/Hermes/iPhone runtime.

Recommendation: Proceed with M0 now. Do not reopen accepted v0 hardening or start an EPUB implementation under this report's read-only authorization.

2. CURRENT COUPLING MAP

Disposition terms: **adapter** = legitimate source-specific behavior; **naming** = safe compatibility debt; **storage/domain/UI** = shared-layer leak; **test** = intentional fixture assumption. "M1" means fix before persisted EPUB support, not in M0.

Assumption	Exact location	Consequence	Disposition
Only fixture and SpaceBattles stories; threadId/sourceUrl required	src/domain/types.ts, StoryMetadata and StoryContentsEntry, lines 42-63	Files cannot express absent URLs or non-forum origin honestly	Domain leak. Add M0 projection; introduce normalized persisted model in M1
Numeric thread/post IDs encoded in strings	src/domain/identity.ts, storyId/chapterId	Broadening one union does not support local identity	Legitimate legacy constructors. Keep exact; add separate UUID constructors in M1
Pending IDs synthesized as story + ":post:" + post	src/storage/repository.ts, listContents, line 79	A file's manifest and body could disagree on chapter ID	Storage constraint. Explicit entry chapter_id in M1
Owner derived by splitting chapter ID	repository.ts, markChapterRead/saveLocator, lines 104 and 315	Opaque IDs can lock the wrong story	Must fix in M1 using persisted chapter-to-story membership
Storage validates numeric same-thread COMPLETE manifest	repository.ts, validateManifest, lines 18-28	File manifests cannot enter this API safely	Adapter validation in shared file. Preserve SB checks; add a narrow generic commit boundary in M1
Primary category required for remote import	importing/import-story.ts, operateStory; repository.ts writers	Missing category is treated as storage failure	Legitimate SB concept. Do not invent EPUB categories
thread_id/post_id/threadmark_entries names	storage/migrations.ts and repository projections	Names imply a forum, but SQL fields are TEXT without numeric constraints	Naming debt. Document opaque file keys; defer renames

Assumption	Exact location	Consequence	Disposition
source_url NOT NULL; primary_category_id nullable	storage/migrations.ts v2	Domain null needs compatibility mapping	Keep SQL empty-string absence, expose null in normalized domain
COMPLETE reconciliation and update timestamps are SB-only	repository.ts, recordUpdateAttempt/reconcileManifest	File import must not impersonate remote refresh	Adapter-specific semantics. Reuse transaction principles, not source API
Body commit reconstructs a numeric chapter ID	repository.ts, commitDownloadedChapter	Arbitrary chapter IDs cannot be committed	Storage constraint. Add validated canonical writer in M1; retain SB wrapper
Duplicate import means resume, never implicit update	importing/import-story.ts, importStory/operateStory	Correct for SB, not automatic file replacement	Keep SB semantics exact; explicit file duplicate/reimport decisions
Network pacing, redirects, access refusal	sources/spacebattles/request.ts, RequestClient; url.ts	Shared global relaxation would weaken accepted protections	Legitimate adapter. Preserve one client, queue and all limits
Thread/post/category DOM extraction	sources/spacebattles/parse.ts; diagnostic.ts	Source-specific discovery and extraction	Legitimate adapter. No generic scraper engine
Same-story ownership and local-write exclusion	importing/ownership.ts; operation finally blocks	A second facade lock would fail; early release reopens stale UI races	Shared invariant. Delegate once; preserve authoritative refresh before release
Only same-origin SB remote images	images/source-url.ts; images/download.ts; images/repository.ts	EPUB package paths are not eligible remote URLs	Legitimate source policy plus storage leak. Separate local asset commit in M1
Shared staged raster files and placeholder rendering	images/files.ts, validate.ts, render.ts; repository.listChapterImages	Core bytes/digest/path machinery already reusable	Preserve; naming SourceImageFiles is harmless temporarily
Pending image slots require source URLs	images/slots.ts; images/repository.ts	Fake URLs would misrepresent local repair	Add EPUB-local slot provenance; downloaded SQL rows already allow null source_url
Only SB management identity; schema must equal 6	storage/management.ts, manageStory, lines 31-45	A v7 migration otherwise breaks existing SB removal	M1 mandatory identity and version-gate update; retain transaction semantics
Cleanup snapshots know only present child tables	management.ts, managementSnapshot and post-COMMIT checks	New sidecars could be orphaned or outcomes misreported	Extend explicit deletion, snapshots and cleanup in same M1 migration slice
Render owner marker only for SB; cleanup parses post suffix	storage/local-files.ts lines 53-55; render-cleanup.ts lines 36-37	File renders lack safe cleanup ownership	M1 deliberate owner-marker extension; preserve strict allowlists and S2A isolation
Whole File.bytes() reads in small-file helper	storage/files.ts, nativeFiles/imageDigest	Reusing it for a large EPUB would buffer the archive	Keep for bounded images; add seekable/chunked package I/O separately

Assumption	Exact location	Consequence	Disposition
Add/Contents invoke SB native services directly	app/add-story.tsx; app/story.tsx, runOperation	Screens know source operations	UI capability leak. M0 guarded facade and eligibility projection
source === "spacebattles" controls maintenance/actions	app/story.tsx, runOperation/confirmManagement/resume controls	Files would inherit wrong or missing controls	M0 replace eligibility predicates; preserve busy/focus/confirmation logic
Management is nested under sourceUrl UI condition	app/story.tsx, maintenance area	A local file with no URL could lose deletion controls	M1 separate local management from attribution block
SpaceBattles/primary-index/download wording	ui/operation-copy.ts; importing/failure-presentation.ts; app screens	File progress/errors would be misleading	Adapter-owned copy/presentation; preserve SB copy in M0
Reader links always open through Linking.openURL	reader/ChapterReader.tsx lines 242-244	Package anchors cannot be passed through unchanged	Explicit initial limitation; typed generic internal links later
Sanitizer removes source IDs and relative links; fixture image sentinel exists	content/sanitize.ts, sanitize/safeLink	EPUB needs preprocessing; source input must not impersonate bundled image	M1 adapter preprocessing. Keep SB canonical algorithm unchanged
Reader template sets lang="en"	reader/template.ts	EPUB language metadata is not represented	Small generic M1 language input with safe default; no style redesign
Continue, Read and availability are mostly origin-neutral	repository.ts continuation/navigation; availability.ts; reader modules	New source can reuse accepted behavior	Preserve; do not infer readability from row existence or Read flags
S1/S2A/fixture assumptions	diagnostics/, app/s1.tsx, app/s2a.tsx, fixtures and tests	Deliberate isolated test paths are not normal sources	Test/diagnostic boundaries. Retain; gate S1 network by audience too
__DEV__ is the only diagnostic build guard	app/index.tsx; sources/spacebattles/native.ts; eas.json	Internal standalone and public builds can both run release JS	M0 explicit audience; public source-module exclusion is a later release gate

All locations are pinned to the reviewed commit. [R2-R12] The historical LOW preference-readback and unsupported-input guidance items were fixed in Pre-S2B; they are not new blockers. [R13]

3. RECOMMENDED ARCHITECTURE

Compare three realistic options

Scores are engineering judgments, not measured results. 5 is most favorable, including lower risk and effort.

Criterion	A: thin registry, retain SB-shaped domain	B: broad source-neutral schema/domain migration	C: separate ingestion, narrow shared contracts
Migration safety	5	2	4
Initial implementation effort	5	2	4
SpaceBattles regression safety	4	2	4
EPUB fit	2	5	5
AO3 fit	3	5	5
Future source fit	2	5	4
Public acquisition gating	4	4	5
Testability	3	4	5
Maintainability	2	4	5

Choose C, implemented incrementally. A is useful as the first facade but insufficient as the final architecture: it leaves fake forum identities and wrong recovery semantics. B moves too many accepted persistence and reader boundaries at once. C preserves the proven reader while allowing different acquisition lifecycles.

The two paths converge after their own validation and preprocessing:

- **Remote pipeline:** recognize and normalize URL, apply source policy, discover a complete source manifest, fetch eligible bodies/assets deliberately, preprocess source markup, sanitize, commit.
- **File pipeline:** select and copy file, validate bounded container and package, build complete spine manifest, validate and stage chapter-at-a-time canonical content/assets, then publish/resume local commits.
- **Official-download specialization:** a format/provenance layer over the file pipeline. It may retain declared source work/chapter references, but does not inherit remote methods or permission.

Responsibility boundaries

Layer	Owns	Does not own
Registry and policy	Recognition routing, implemented operations, static distribution decision, descriptors	Network transport, parser internals, database writes
Remote adapter	Site IDs/URLs, source manifest rules, DOM extraction, per-source access and asset policy	Reader rendering, shared locators, global UI state
File importer	Container/path/XML validation, spine, local resources, import-version mapping	Network fetch, login, remote refresh
Orchestrator	Explicit operation lifecycle, cancellation, staged completeness, commit order, progress	Alternate canonical HTML formats or second owner around legacy service
Canonical content/asset layer	Accepted sanitization, deterministic blocks/hashes, static raster validation and generated local references	Site-specific permission or remote host broadening

Layer	Owns	Does not own
Repository	Relationships, availability, Read/locator durability, atomic reconciliation/commit and cleanup contracts	URL recognition or best-effort content extraction
Screens and presentation	Capability-driven actions, source labels, progress/error wording, current race controls	Source-specific branches or acquisition on opening
Reader	One available canonical chapter, trusted bridge, restoration and quiesced navigation	Source lookup, repairs, network, raw EPUB CSS/HTML

For M0, wrap the existing import/update/image/native management entry points. Do not move the entire SB pipeline into a new directory merely to match this diagram. A later extraction may isolate source validation from generic persistence when EPUB actually needs a second writer.

4. ADAPTER / IMPORT CONTRACT

The following sketches describe the target boundary. **M0 implements only the pure origin/capability projection, fixture/SB descriptors, policy decision and guarded legacy service composition.** It does not add these file interfaces to production as pretend support.

Identity, origin and display model

```
type StoryId = string;
type ChapterId = string;
type RemoteSite = 'spacebattles' | 'ao3' | 'royalroad' | 'fanfiction';
type FileFormat = 'epub' | 'html';
type HttpsUrl = string; // constructed only by validated URL helpers

type Origin =
  | { kind: 'fixture'; key: string }
  | { kind: 'remote'; adapterId: RemoteSite;
      workId: string; canonicalUrl: HttpsUrl }
  | { kind: 'file'; importerId: FileFormat;
      importId: string; packageId: string;
      declaredSource: DeclaredSource | null };

interface DeclaredSource {
  site: RemoteSite | null;
  workId: string | null;
  url: HttpsUrl | null;
  basis: 'package-metadata' | 'recognized-export';
  // A declaration is not authentication, ownership or live-adapter approval.
}

interface StoryCore {
  id: StoryId;
  origin: Origin;
  title: string;
  author: string | null;
  sourceUrl: HttpsUrl | null;
}
```

```
interface OptionalMetadata {
  creators?: readonly string[];
  description?: string; language?: string; rights?: string;
  identifiers?: readonly string[];
  publishedAt?: string; modifiedAt?: string;
  completion?: 'complete' | 'in-progress' | 'unknown';
  rating?: { scheme: string; value: string; declared: true };
  warnings?: readonly string[];
  fandoms?: readonly string[];
  tags?: readonly string[];
}

interface ManifestEntry {
  chapterId: ChapterId;
  resourceKey: string; // stable key within the stored import/work instance
  title: string;
  position: number;
  sourceUrl: HttpsUrl | null;
  role: 'primary' | 'auxiliary';
}

interface CompleteManifest {
  state: 'complete';
  storyId: StoryId;
  entries: readonly ManifestEntry[];
  revision:
    | { kind: 'remote-observation'; observedAt: number }
    | { kind: 'package'; packageId: string; sha256: string };
}
```

Core identity/title/order/body availability remain structured and queryable. Optional metadata is bounded plain text. M1 persists title, creators, language, identifier, rights, description, dates and declared source attribution where supplied. Ratings/warnings/fandom/tags may be retained as bounded optional metadata when an importer can identify their meaning; they do not require Library filters or dozens of new UI rows. Unknown completion remains unknown. Import time and source publication/update time are distinct.

Multiple creators are preserved in optional metadata; the existing author field receives a deterministic bounded display string. Author notes remain in the canonical body where they occur. Covers are optional validated images, not an extra reader or identity source.

Separate acquisition interfaces

```
interface OperationContext {
  signal: AbortSignal;
  report: (event: IngestProgress) => void;
}
// IngestProgress is a discriminated remote/file phase union.
// Existing SB progress/result/error types remain unchanged in M0.

interface RemoteAdapter {
  descriptor: RemoteDescriptor;
  recognizeUrl(value: string): RemoteReference | null;
  loadManifest(ref: RemoteReference, ctx: OperationContext):
    Promise<CompleteManifest>;
  loadChapter(ref: RemoteReference, entry: ManifestEntry,
    ctx: OperationContext): Promise<PreparedInput>;
  loadAsset(ref: RemoteAsset, ctx: OperationContext):
    AsyncIterable<Uint8Array>;
}
```

```
interface FileImporter {
  descriptor: FileDescriptor;
  inspect(file: SelectedFile, ctx: OperationContext):
    Promise<ValidatedPackage>;
  stage(pkg: ValidatedPackage, ctx: OperationContext):
    AsyncIterable<StagedChapter>;
}

type AssetReference =
  | { kind: 'remote'; adapterId: RemoteSite; url: HttpsUrl }
  | { kind: 'package'; packageId: string; resourcePath: string };

// PreparedInput contains normalized body markup and out-of-band image refs.
// StagedChapter contains existing canonical HTML/hash/version/blockIds/links,
// plus validated generated-file refs. Neither exposes source scripts to Reader.
```

RemoteReference, SelectedFile, ValidatedPackage, RemoteAsset and StagedChapter are validated application values, not unchecked strings. Their constructors belong to the responsible adapter/importer and are not generic public parser APIs. Neither interface writes the database. File iteration is sequential and bounded; do not collect every yielded body into an array.

Do not implement a file loadAsset method by passing an archive path to an HTTPS downloader. The local importer reads virtual ZIP members, validates bytes and stages generated local files. Share the accepted I2 validation/staging/rendering primitives, not its SB URL policy or network discovery.

Typed capabilities and policy

```
type DistributionStatus =
  | 'internal-only'
  | 'public-approved'
  | 'public-file-import-only'
  | 'disabled';

type UpdateSupport =
  | { mode: 'none' }
  | { mode: 'complete-manifest-refresh';
      removedEntries: 'retain-saved-unlisted' }
  | { mode: 'explicit-reimport';
      target: 'explicit-existing-import';
      matching: 'conservative-resource-map' }
  | { mode: 'replace-whole-document'; target: 'explicit-existing-import' };

type BodyRecovery =
  | { kind: 'bundled' }
  | { kind: 'remote-refetch' }
  | { kind: 'retained-package'; packageId: string }
  | { kind: 'select-original'; expectedSha256: string }
  | { kind: 'unavailable' };

interface PolicyEvidence {
  status: DistributionStatus;
  scope: 'direct-acquisition' | 'user-selected-file';
  evidenceRefs: readonly string[];
  reviewedAt: string;
  reviewDueAt: string | null;
  authorizationExpiresAt: string | null;
}
```

```
interface RemoteDescriptor {
  kind: 'remote';
  id: RemoteSite;
  label: string;
  policy: PolicyEvidence;
  updates: UpdateSupport;
  assets: { kind: 'allowed-origins';
    origins: readonly HttpsUrl[]; redirectPolicyId: string };
}
interface FileDescriptor {
  kind: 'file';
  id: FileFormat;
  label: string;
  policy: PolicyEvidence;
  assets: { kind: 'package-only' };
}
```

Recovery is derived from the stored item and current package availability, not frozen in the source descriptor. Likewise, an update mode is advertised only once its implementation is accepted. M1 initially supports a new import, exact-package duplicate handling and deliberate local resume/restore; changed-package replacement of an existing item remains disabled until its matching/reconciliation slice after backup.

A registry should return typed action descriptors such as refresh-manifest, resume-remote, restore-package or select-original, including presentation keys and unavailable reasons. Screens combine those with existing busy/refreshing/confirmation state. They must not replace a compound race guard with a static capability bit. The service rechecks eligibility; hiding a button is not enforcement.

Failure taxonomy

Preserve all allowed-path S1Failure/ImportFailure values and copy in M0. Add a small facade rejection type with unsupported-source and policy-disabled, plus an unsupported-operation reason where necessary. Map it at the presentation boundary without catching and relabeling legacy failures.

For M1, use a tagged failure with stage, stable code, safe context, retry action and whether any canonical commits exist. Required codes: unsupported-file-type, malformed-archive, unsupported-encryption, incomplete-manifest, malformed-xml, unsupported-structure, resource-limit, missing-original, stale-package, storage, cancelled and update-unsupported. Retain access-refused/rate or network failures only on remote paths.

Messages distinguish "No story was added", "Validated sections already saved are retained", and "Could not confirm the final local state". Storage exceptions after COMMIT must use authoritative readback, as current management does. Do not claim rollback merely because a promise rejected. Never expose real prose, local provider URLs, credentials or full parser excerpts in logs. [R4-R6]

5. IDENTITY DECISIONS

Origin	Story identity	Chapter identity and limits
Existing SpaceBattles	Exactly spacebattles:thread:<normalized numeric thread ID>	Exactly existing story ID + :post:<normalized numeric post ID>. Never regenerate old values
Arbitrary EPUB	"file:" + a persisted random UUID for the import instance	"file:" + import UUID + ":chapter:" + persisted chapter UUID; map to normalized package resource path
Generic HTML	New local import UUID	One persistent chapter UUID initially; explicit whole-document replacement later
Future direct AO3	Deterministic adapter namespace + verified work ID, once source contract is implemented	Actual durable chapter ID/reference when supplied and validated. Never pretend a chapter ordinal or generic anchor is a durable AO3 ID
AO3 EPUB/HTML file	Local import UUID, with separately retained declared source work reference	Local chapter UUIDs; use export-provided references only as scoped matching evidence
Future Royal Road	Deterministic namespace + verified fiction ID, subject to approved API/adapter contract	Verified chapter IDs; do not promise a contract from uninspected live pages
Future FanFiction.net	Deterministic namespace + verified story ID, subject to authorization	A chapter ordinal is weaker than a durable ID. Treat boundary/reorder uncertainty explicitly

These future source rules are design recommendations, not claims that live AO3/RR/FFN adapters have been tested.

Package fingerprint: Compute incremental SHA-256 over the exact retained bytes. Identical bytes identify an exact package version; ZIP repacking can change that fingerprint without changing the work. It is not a logical story ID, author identity, permission proof or content authenticity signature.

Duplicate import: For an exact known package, show the existing local item and offer explicit open/continue. A user-requested second copy receives a new import UUID and independent Read/locator state. Do not silently duplicate or merge. Concurrent imports of the same hash must resolve the duplicate choice before canonical publication.

Changed file: The user explicitly chooses an existing import as the target, or adds a new copy. Similar title/author/identifier/declared URL only suggests a possible match. It cannot authorize replacing an item. M1 defers changed-package reconciliation; later work requires backup first.

Chapter stability: Persist UUID-to-resource mapping. Renames and reorder alone preserve identity when the normalized resource mapping is unique. A resource path is evidence scoped to an import, not globally durable identity. OPF item IDs, content digests and headings may help review ambiguity but must not silently override it. A byte-identical resource can corroborate a moved path; ambiguous matches require user choice or new IDs.

Split/merge or conflicting map: Allocate new chapter IDs; do not spread one Read flag or locator across unrelated replacements. Preserve old state and readable old content as explicitly previous-import sections if that feature is implemented. Otherwise reject in-place replacement and offer a separate import. Changed hash retains the current progression fallback, not an edit-proof passage guarantee.

Multiple origins/formats: Allow independent local instances with optional known/declared external work references. Do not build a many-to-many work/edition synchronization system now. An EPUB and a future approved remote version may coexist. No automatic cross-origin Read-state merge.

Nullable URLs: The normalized model exposes null for no applicable source URL. Existing SQLite empty strings remain a temporary absence representation. Never synthesize a network URL to satisfy a legacy field.

6. STORAGE / MIGRATION PLAN

What is needed when?

M0: no migration. Add a source-neutral view/projection over the existing StoryMetadata shape. Keep current schema v6, repository writers, IDs and accepted fixtures unchanged.

M1a format/runtime proof: no normal-library migration. Exercise synthetic EPUBs in operation-owned scratch/staging or an isolated test harness. Do not use or reset S2A's dedicated database.

M1 persisted EPUB: a bounded v7 migration is justified. Keep stories and chapters, their payloads and foreign-key targets. Rebuild only the manifest leaf table to add explicit chapter_id, then add two file-specific sidecars.

Minimum proposed schema

Storage area	Decision
stories	Keep existing columns. For source='epub', thread_id is the real import UUID storage key; source_url='' maps to domain null; category is null
chapters	Keep existing columns. post_id is the persisted opaque local entry key. Existing id is the canonical chapter ID
threadmark_entries	Keep name and all current fields; add chapter_id TEXT NOT NULL UNIQUE; use it in current-schema projections
local_imports	One row per local story: retained package ID/path/hash/bytes, importer version, selected OPF path, import time and bounded optional metadata
local_import_entries	One row per local manifest entry: normalized resource path, OPF item ID if present, raw resource hash and primary/auxiliary role
chapter_images	Reuse existing nullable source_url and validated local-file fields; add no generalized media schema
reading_positions/settings/assets	Preserve exactly; no resanitization or inference of Read state

Using the legacy TEXT fields is an explicit compatibility mapping. SQL does not require numeric thread/post values or restrict source to the current TypeScript union. Do not pass file keys through legacy numeric identity constructors. Optional source metadata JSON is capped and versioned; identity, order, Read flags, availability, package paths and relationships stay in structured columns. [R2-R4]

The entry sidecar avoids storing the whole manifest in JSON and supplies the resource map needed for exact-package resume and deterministic local image rediscovery. Before resume or repair, verify both the retained archive fingerprint and the recorded resource hash. The importer version must name an

actually supported deterministic extraction/preprocessing recipe, including its sanitizer profile; never update that version in place merely to run newer code.

No separate local-image mapping table is needed initially: reparse the pinned original chapter with the recorded importer version, reproduce image slots, and require exact canonical HTML/hash/version/block/link compatibility before attaching repaired image bytes. This deliberately reuses I2's conservative compatibility principle. The immutable package and pinned recipe establish image provenance: canonical placeholder text alone does not encode the source image. Match reproduced slot ordinals and block IDs to the saved descriptors as well as checking canonical equality. If that cannot be reproduced, show stale-package/unsupported-import-version; never change saved prose as an image repair.

Body restoration is a separate operation. Remove Reading Copy intentionally clears old canonical fields, so restoration must not require equality with those empty fields. Retain both sidecars, reproduce from the verified original and supported recipe, and refill the same unavailable shells while preserving Read state and locators. Exact saved-body equality applies to image-only repair while a body is available.

Illustrative SQL, not an implementation patch

Execute through the existing serialized exclusive migration transaction. Validate old invariants before copying; if they fail, preserve v6 and report a storage failure rather than repairing silently.

```
CREATE TABLE threadmark_entries_v7 (
  story_id TEXT NOT NULL REFERENCES stories(id),
  post_id TEXT NOT NULL,
  chapter_id TEXT NOT NULL UNIQUE,
  title TEXT NOT NULL,
  position INTEGER NOT NULL CHECK (position >= 0),
  source_url TEXT NOT NULL,
  currently_listed INTEGER NOT NULL DEFAULT 1
    CHECK (currently_listed IN (0,1)),
  is_read INTEGER NOT NULL DEFAULT 0 CHECK (is_read IN (0,1)),
  PRIMARY KEY (story_id, post_id)
);

INSERT INTO threadmark_entries_v7
  (story_id, post_id, chapter_id, title, position,
   source_url, currently_listed, is_read)
SELECT e.story_id, e.post_id,
       COALESCE(c.id, e.story_id || ':post:' || e.post_id),
       e.title, e.position, e.source_url,
       e.currently_listed, e.is_read
FROM threadmark_entries e
LEFT JOIN chapters c
  ON c.story_id = e.story_id AND c.post_id = e.post_id;

DROP TABLE threadmark_entries;
ALTER TABLE threadmark_entries_v7 RENAME TO threadmark_entries;
CREATE UNIQUE INDEX current_threadmark_positions
  ON threadmark_entries(story_id, position)
  WHERE currently_listed = 1;
```

There is deliberately **no foreign key from manifest chapter_id to chapters**: a complete manifest includes pending chapters before a body row exists. The migration preserves actual stored chapter IDs when present and uses the accepted legacy constructor only for pending legacy entries. Current-schema code must then require explicit chapter_id, rather than fabricate a missing mapping.

```
CREATE TABLE local_imports (
  story_id TEXT NOT NULL PRIMARY KEY REFERENCES stories(id),
  format TEXT NOT NULL CHECK (format = 'epub'),
  package_id TEXT NOT NULL UNIQUE,
  package_relative_path TEXT,
  package_sha256 TEXT NOT NULL CHECK (length(package_sha256) = 64),
  package_bytes INTEGER NOT NULL CHECK (package_bytes > 0),
  importer_version TEXT NOT NULL,
  opf_path TEXT NOT NULL,
  imported_at INTEGER NOT NULL,
  metadata_json TEXT NOT NULL DEFAULT '{}'
    CHECK (length(CAST(metadata_json AS BLOB)) <= 65536)
);

CREATE TABLE local_import_entries (
  story_id TEXT NOT NULL,
  post_id TEXT NOT NULL,
  resource_path TEXT NOT NULL,
  opf_item_id TEXT,
  resource_sha256 TEXT NOT NULL CHECK (length(resource_sha256) = 64),
  role TEXT NOT NULL CHECK (role IN ('primary','auxiliary')),
  PRIMARY KEY (story_id, post_id),
  UNIQUE (story_id, resource_path),
  FOREIGN KEY (story_id) REFERENCES local_imports(story_id),
  FOREIGN KEY (story_id, post_id)
    REFERENCES threadmark_entries(story_id, post_id)
);

PRAGMA user_version = 7;
```

The optional package path allows an honestly missing original while retaining fingerprint/provenance. M1 initially creates a generated non-null path. Validate lower-case hex hashes, generated path grammar, JSON object shape and all bounds in application code; length checks alone do not validate a hash or JSON. Exact byte duplicate packages across different stories remain allowed; do not create a UNIQUE hash constraint.

The unique resource path encodes the initial rule that repeated spine references to the same resource are unsupported. A future importer that accepts repeated occurrences must deliberately version that mapping, not reuse this schema assumption silently.

Migration and failure sequence

1. Finish the accepted v1-v6 migration path and verify the database is the normal library.
2. In one exclusive transaction, create/copy the leaf table, add sidecars, recreate current-only uniqueness, verify counts/identity/FKs, then set user_version=7.
3. Roll back all schema/version/data changes on failure. A retry starts from intact v6. Never run resanitization or rewrite old chapter payloads.
4. Update latest-schema fixture seeding, initial/reconciled manifest inserts and projections together. Maintain the real v1 S2A compatibility path.
5. In the same slice, replace parsed ownership with validated stored membership and update management's explicit schema==6 gate to the supported normal schema.
6. Add explicit sidecar deletion and snapshot/post-COMMIT checks. Capture package paths before DB deletion; clean only generated, unreferenced, inactive files after commit.
7. Preserve existing shell/availability and Read/locator semantics. A failed cleanup is logical deletion success with an explicit cleanup warning, not reference resurrection.

Current exclusive transactions may use a fresh connection where enabling FKs inside BEGIN cannot take effect; the existing global foreign_key_check detects invalid references but does not perform cascades. Therefore new sidecars must participate in explicit deletion. [R3-R4, E17]

SQLite's current documentation includes ALTER COLUMN SET/DROP NOT NULL from 3.53.0. The native bundled SQLite version was not verified, and this proposal does not depend on that newer syntax. A narrow leaf-table rebuild follows the project's established pattern. [E17]

Retained files and durability

Store packages under a generated app-private namespace such as imports/<UUID>.epub, outside the Reader read-access directory. Never use an archive member name as a disk pathname. Stage and fingerprint the copy before committing its reference; never overwrite a referenced package. Treat database reference publication as the boundary between an active stage and retained content.

A crash before publication leaves a recoverable or removable stage. Publish the complete manifest, retained-package reference and every entry recipe in one initial transaction. A crash during later body publication leaves that complete recovery recipe and only valid committed bodies; explicit Continue Import may resume. No automatic acquisition, import or cleanup on opening a story.

Full deletion removes the selected story's sidecars, entries, bodies, images and locators; preferences, other stories, fixture assets and S2A remain. Reference-aware cleanup protects active stages and shared image files. M1 does not deduplicate original package storage across separate copies, avoiding a package reference-count subsystem.

7. EPUB SECURITY MODEL

Supported profile and standards boundary

Use EPUB 3.3 as the stable baseline and official legacy EPUB2 OPF/OPS documentation for EPUB2. EPUB3 reading systems have requirements beyond this proposal, including ZIP64 support; Stowaway's initial ZIP32-only, UTF-8-only import profile is deliberately narrower. Do not advertise it as a fully conforming EPUB renderer. [E1-E5]

Recommended M1 profile: reflowable EPUB2/3 containing local XHTML prose; stored or raw-DEFLATE ZIP32 members; OPF metadata/manifest/spine; EPUB3 nav or EPUB2 NCX labels; supported static raster images; Stowaway-generated canonical content. Fixed-layout dependence, DTBook as primary content, media overlays, scripted publications and unsupported primary media fail with an actionable unsupported-structure result.

The container points to the package document; manifest resources and spine references define the selected publication, while nav/NCX provide navigation. That separation is the reason to use spine items for boundaries and navigation entries for labels. [E1, E3-E4]

Exact initial limits

These are **proposed Stowaway acceptance limits**, not EPUB-standard limits or measured iPhone guarantees. MiB means 1,048,576 bytes. Existing sanitizer and individual-image limits remain unchanged.

Quantity	Initial limit and accounting
Compressed selected EPUB	100 MiB
Indexed ZIP members	10,000, including ignored entries
Central directory	4 MiB
Declared uncompressed total	256 MiB across every indexed member
Actual output during one preflight	256 MiB; count all produced bytes, including later-rejected resources and repeated work
Spine items	2,000
File input/inflater output buffer	64 KiB each; no whole-archive buffer
container.xml and encryption.xml	65,536 bytes each
OPF, nav and NCX XML	1,000,000 decompressed bytes and 1,000,000 JS string units each
Raw XHTML	4,000,000 decompressed bytes; decoded source <=1,000,000 JS string units before XML parsing
Rewritten sanitizer input	source.length <=1,000,000 before calling the accepted sanitizer
XML accepted complexity	Depth 64; 30,000 total nodes; 64 attributes per element
Path	1,024 UTF-8 bytes; each segment <=255 bytes
Canonical staged text	64 MiB total UTF-8 bytes; account for any duplicated persisted representation
Optional metadata	65,536 UTF-8 bytes per story; title/creator display <=512 characters; URL <=2,048; language <=64; description/rights <=8,192 each
Accepted unique images	2,000 assets and 64 MiB bytes total after validated deduplication
Individual static raster	Existing 2,000,000-byte, 2048-per-dimension and 4,194,304-pixel limits
Per-chapter displayed images	32 instances and 8,388,608 aggregate intrinsic pixels; count repeated occurrences of a shared asset each time
Expansion-ratio early filter	Reject >200:1 for members declaring >1 MiB output; actual byte/allocation limits remain authoritative
Free-space starting reserve	At least 512 MiB available before selection/copy; estimate simultaneous copies and handle write failure throughout
Runtime feasibility targets	No >250 ms uninterrupted processing stall in the adversarial corpus; cancellation observed within 1 s outside an OS picker/provider call

The last two rows are conservative initial engineering gates, not Apple device guarantees. The 100 MiB archive limit is an acceptance cap: the iOS provider/picker may create a temporary copy before JavaScript can inspect its length. Reject an oversized selection as soon as size is available; do not claim this bounds the provider's preceding copy, and clean only app-owned files. The disk estimate includes a provider temporary copy, app-owned package, canonical text, accepted images, SQLite/WAL and a reserve; an initial free-space check cannot guarantee subsequent writes. Do not increase caps to pass a fixture.

XML caveat: The proposed parse-xml library builds a tree synchronously. Depth/node/attribute checks are acceptance checks after parsing unless an independently validated early scanner is added. They do

not currently prevent the parser's initial allocation. Enforce the decoded source cap first; measure worst-case densely nested/tiny-node/long-attribute/malformed-final-token input at that cap. A timeout cannot interrupt synchronous parsing. If the device memory or stall target fails, change the parser strategy and re-review M1a before enabling import.

Image caveat: 8,388,608 RGBA pixels correspond to about 32 MiB for one decoded representation; caches/copies/WebView overhead are additional. This is not a total memory guarantee. Use the current PNG/JPEG/WebP structural validation and its animation rejection. Count every image occurrence, not just every unique file. [R7-R8]

Archive validation and virtual paths

The recommended ZIP reader is a small seekable ZIP32 index reader using bounded raw inflation. The ZIP format specifies central and local headers, descriptors, compression methods and integrity fields; these must agree rather than being trusted independently. [E6]

1. Check file length first. Seek through a bounded EOCD search window (at most 65,557 bytes), then validate the bounded central directory and every indexed member range.
2. Require a valid EPUB mimetype member and a valid META-INF/container.xml. Select exactly one supported rootfile; reject ambiguous multiple supported renditions in the first profile.
3. Reject ZIP64, split archives, ZIP encryption, unsupported compression, symlinks, device/special entries, impossible offsets, overlaps, truncated metadata and contradictory local/central flags or names. Permit normalized zero-payload directory entries as ignored structural members, count them toward entry limits and never materialize them. Support ordinary ZIP32 data descriptors deliberately; verify their consistency.
4. Reject duplicate normalized names, absolute paths, drive paths, backslashes, NUL/control characters, member names containing traversal segments and ambiguous name encoding. Do not accept a parser's convenient overwrite-last behavior.
5. Resolve document-relative links within a virtual package root. "../Images/picture.jpg" is allowed if it remains inside that root; it is not the same as an archive member named "../escape". Decode once under a defined URL policy; reject encoded separators, escapes, conflicting aliases and double-decoding ambiguity.
6. Read only needed members by offset, using caller-owned fixed output buffers. Check remaining budget before the next output write; detect excess output using at most remaining+1 bytes. Require progress, valid termination, expected compressed consumption, actual length and CRC32 agreement.
7. CRC is verified for every member actually read. Never claim validation of ignored members that were not inflated. All indexed paths/ranges still undergo structural checks.

Optional oversize or unsupported images may be skipped before inflation and represented by predetermined placeholders. But a selected member that begins inflation and reveals truncation, CRC/length mismatch or contradictory structure is a fatal package-integrity failure, not an ordinary image omission.

Reject, strip, transform and preserve

Treatment	Exact initial behavior
Reject package	Invalid container/OPF/required XHTML, broken spine references, conflicting IDs/paths, duplicate spine resource references, unsupported required media, malformed XML, prohibited entity definitions, invalid encoding, required package/XML/text limit breach, fatal archive-integrity failure or meaningful encryption. Optional-image limit exceptions below remain placeholders

Treatment	Exact initial behavior
Reject as unsupported encryption	ZIP encryption; encrypted reading content/assets; unknown encryption algorithm, key material or unexpected transforms; invalid encryption references
Allow narrow font-obfuscation declaration	Only recognized IDPF font obfuscation affecting resources proven to be discarded fonts; no alias to a spine/cover/accepted asset; every declaration validated; no deobfuscation performed
Strip active/source presentation	Scripts, event handlers, forms, iframe/object/embed, meta refresh, base/xml:base behavior, stylesheets, style attributes/elements, CSS url(), remote fonts, source processing instructions and active remote resources
Strip unsupported image sources	Remote/protocol-relative/data/blob/file/javascript URLs; no fallback network fetch; local SVG/MathML are not rendered
Transform content	Extract XHTML body only; retain supported text, headings, emphasis, lists, tables, quotes, code and notes through the existing allowlist; normalize adapter-specific wrappers before sanitization
Transform images	Resolve ZIP-local raster references, reset every source img src including fixture:// lookalikes, assign out-of-band canonical image slots and generated filenames
Preserve metadata	Bounded plain text title/creators/language/identifier/rights/description/dates/declared origin; never active HTML
Preserve external links	Only existing validated HTTPS-link policy, opened after explicit user action
Internal links in initial M1	Preserve link text; remove operational relative/fragment hrefs. Show an import limitation notice. No archive file:// navigation
Missing/rejected optional raster	Stable placeholder plus reason; no background repair. An archive-integrity error remains fatal
Empty XHTML spine section	Preserve its manifest position with an app-generated empty-section notice and import warning; do not silently renumber or merge
Empty result caused by unsupported primary content	Reject unsupported-structure rather than pretending a meaningful section imported

encryption.xml is not synonymous with DRM; font obfuscation also uses it. The narrow allowance above is a discard-only decision, not a decryption capability. [E1] Metadata rights.xml or a rights statement alone does not prove encryption; validate actual mechanisms and never implement circumvention.

Decode UTF-8 strictly and reject invalid bytes or contradictory encoding declarations. UTF-16 input is unsupported in this first profile. Never fetch a DTD, XInclude or external entity. Accept no internal subset or ENTITY declaration. A recognized legacy XHTML external DOCTYPE may be discarded without retrieval; otherwise reject unsupported DTD forms. Support XML's predefined/numeric entities; unsupported named XHTML entities produce a clear compatibility error unless a fixed reviewed mapping is added in a separately tested importer revision. [E4, E13]

Deterministic chapter boundaries

One Stowaway chapter per supported XHTML spine item, in spine order. Preserve front matter, end matter and author notes. Include linear="no" spine items at their declared positions, marked auxiliary in import metadata. This chooses preservation over a separate supplemental-reader UI; it is a Stowaway import rule.

Do not split on headings or nav entries, and do not combine tiny items. Those approaches turn formatting changes into identity changes. Nested TOC entries can help label the containing spine item, but do not reorder or multiply it. Choose the first suitable document-level nav/NCX label, then first heading/document title, then "Section N". Duplicate titles are fine.

Reject repeated references to the same spine resource in the initial profile, since occurrence identity is otherwise ambiguous. Nav-only XHTML outside the spine is not imported initially; links to it remain unavailable and that limitation is disclosed. Do not classify auxiliary or nav-only content as remote "saved-unlisted" chapters.

Later local links: A separate generic content/reader slice may introduce a typed link target {kind:'chapter', chapterId, blockId|null}. Build source-anchor-to-generated-block maps during the EPUB-specific preprocessing profile; require actual target existence, keep unknown anchors inert, and route cross-chapter navigation through the accepted quiescence handshake. Preserve the legacy string HTTPS-link representation and SB sanitizer output. No source-specific Reader renderer or package WebView is justified.

Immutable staging and cancellation

Before any normal-library publication, create a complete validated staging record binding the operation/import ID, immutable package SHA-256, importer/sanitizer version, complete spine plan, canonical chapter artifacts, accepted asset hashes and predetermined omissions. Stage chapter by chapter; keep only one XML/body and a bounded asset window in memory.

Promotion consumes that record. It must not reparse arbitrary source content or make new resource decisions while writing canonical rows. Write and verify final generated files before committing DB references. The manifest/package/entry-recipe transaction precedes body publication and is atomic. A chapter transaction publishes only a complete canonical chapter unit and its known asset refs/placeholders.

Cancellation during copy, hash, package validation or canonical staging publishes no new story. Cancellation after the full stage is ready but before publication also publishes none. Cancellation during publication retains only completed chapter transactions, the full original manifest/ordinals and retained package. Explicit Continue Import recreates and validates missing stages as necessary. No partially written body is readable.

On restart, a stored package path is not proof that a file exists or still hashes correctly. Revalidate before repair/resume. Cleanup removes only owned, unreferenced staging files and never races an active import. No orphan sweep over arbitrary app storage.

8. UPDATE / CAPABILITY MODEL

Keep source support, implemented operations, distribution permission and current item state separate. The registry describes potential operations; service eligibility combines those with the build audience and current stored item. The screen adds its accepted busy/focus/refresh constraints.

Item and state	Supported update/recovery	UI behavior
Internal SpaceBattles, complete	Deliberate complete-manifest refresh; resume only if bodies missing	Existing Check for Updates and Resume Download semantics
SB manifest incomplete/refused	No successful reconciliation	Retain previous complete contents and success timestamp; show typed failure

Item and state	Supported update/recovery	UI behavior
New EPUB	No network updates; explicit local processing	Import File, then progress through validation/staging/saving
Partial EPUB, original retained and importer supported	Continue missing local commits	Continue Import; no source requests
EPUB reading copy removed, original usable	Rebuild from exact package	Restore Reading Copy
EPUB original missing, bodies present	Read existing canonical copy; no automatic repair	Reading remains available; original-required actions explain selection
EPUB original missing, bodies absent	Explicitly select identical original package	Select Original File; verify SHA before restoring same identities
Different package selected	Not exact recovery	Stale Package; offer a new import. In-place reimport remains deferred
Future accepted changed-package workflow	Explicit targeted reconciliation using conservative map	Reimport File, preview identity changes and preserve unmatched old state
Fixture	Bundled reading only	No source acquisition, removal or unsupported update controls
Public/unconfigured build with SB item	Local reading/Read/locator/removal remain available; acquisition denied	No acquisition actions; service rejects a stale/deep-linked attempt

SpaceBattles remains exact: complete manifest before reconciliation; stable post IDs; metadata-only rename/reorder; additions begin Unread; removals remain saved-unlisted where bodies exist; reappearance reuses old bodies/Read/locators. Duplicate import stays resume-only and a complete duplicate performs no HTTP. Attempt time and successful-check time remain distinct. [R3, R5-R6]

EPUB Remove Downloads decision: Retain the original archive. In the first persisted text proof, keep local purge hidden until exact-package restoration, interruptions and missing-original handling pass. By M1 acceptance, expose "Remove Reading Copy" only when a usable retained package and supported importer can restore it. Confirmation states that the original remains and storage reclaimed excludes it. Full Remove Story remains available independently.

Removing a reading copy keeps manifest identities, Read state and locator-bearing shells. Continue Reading excludes unavailable bodies; it becomes eligible again after the same body identity is restored. Never reconstruct continuation merely from a Read flag. If an original is unexpectedly lost, surviving canonical chapters remain readable. Selecting the exact original can repopulate shells without resetting Read/locators; a changed file cannot silently fill them. [R3-R4, R9]

Future AO3: File versions are package snapshots. A later approved direct adapter may use stable work/chapter references, complete manifest observations, and explicit updates; missing/deleted/restricted responses must not masquerade as a complete empty work. Rating/warning changes are metadata changes, not necessarily new chapter identities. Any distribution content gate is re-evaluated before newly acquired material is committed. Preserve local state according to the documented product policy; do not silently erase it on an access refusal.

No automatic recovery: There is no source work on screen open, startup, foregrounding, source-label inspection or Reader image failure. File repair and remote refresh are deliberate operations. No new global state framework, background job service or universal scheduler.

9. PUBLIC DISTRIBUTION GATING

Current source posture, verified against official sources

This table preserves the repository's working permission baseline. It describes acquisition paths, not a legal conclusion about every imported book. [R14]

Acquisition path	Registry posture now	Public consequence
Generic user-selected EPUB	public-file-import-only, implementation pending	Preferred foundation once parser/security/product gates pass
Generic local HTML	public-file-import-only, implementation deferred	No live-source fetching implied
AO3 official download selected by user	public-file-import-only, specialization pending	Goes through file importer; preserve attribution and format limits
AO3 direct integration	disabled pending specific approval/content decision	Conditional candidate, not an enabled adapter
SpaceBattles direct	internal-only	Preserve accepted personal/internal scope; public permission pending
Royal Road direct	disabled	Require express permission or an official approved integration route
FanFiction.net direct	disabled	Require express written authorization before public direct integration

Royal Road's official terms expressly restrict automated access, scraping/caching and bots unless permitted. FanFiction.net restricts access through technology other than the Service and outside its normal functionality, separately restricts automated systems exceeding ordinary browser request volume, and requires authorization for distribution of Service content. Those current texts support retaining the existing cautious registry posture. [E22-E23]

AO3's official FAQ permits responsible automated access subject to constraints; that is not general approval for an App Store integration. The official OTW article explains that AO3 has no official app and that download facilities support offline reading. Declared or recognized AO3 file provenance still does not confer author rights or direct-source permission. [E19-E21]

Policy-report refinement: The repository mentions likely excluding Explicit/Not Rated for a future AO3 public path. AO3's FAQ places Mature within the adult-content range and allows explicit content there. If the later product goal is to exclude explicit material, Mature must also be considered; a rating filter alone is not a content guarantee. This is a correction to the assumption, not evidence that AO3 policy changed after the repository report. It does not justify adding moderation/discovery work to M0. [R14, E20]

SpaceBattles' official terms URL was discoverable, but direct retrieval returned access failure during this review; no fresh robots-policy verification was established. Do not infer a permission change or claim written approval. Preserve the accepted personal/internal posture and the pending public-permission requirement. No live stories were fetched and no permission outreach was sent. [R14, E18]

Apple's current Review Guidelines include third-party-service permission obligations, restrictions on hidden/dormant functionality and application of the guidelines to TestFlight. A generic file importer lowers direct-source integration risk; it does not by itself establish complete App Store readiness. [E24]

M0: explicit audience and guarded dispatch

Use **build audience**, not EAS Update "channel", as the concept:

```
type BuildAudience = 'internal-personal' | 'public' | 'unconfigured';

function parseBuildAudience(value: unknown): BuildAudience {
  return value === 'internal-personal' || value === 'public'
    ? value : 'unconfigured';
}

const BUILD_AUDIENCE = parseBuildAudience(
  process.env.EXPO_PUBLIC_STOWAWAY_BUILD_AUDIENCE
);
```

Use a statically referenced environment value embedded by Expo. Missing/malformed values deny source acquisition; they do not fall back to internal. Preserve ordinary internal development commands using a small cross-platform Node launcher that supplies the value deliberately, without adding a dependency. Do not infer permission from __DEV__, NODE_ENV, profile name at runtime, package provenance or a user setting. Expo documents EXPO_PUBLIC substitution and EAS profile environment values. [E10]

```
{
  "build": {
    "development": {
      "developmentClient": true,
      "distribution": "internal",
      "env": {
        "EXPO_PUBLIC_STOWAWAY_BUILD_AUDIENCE": "internal-personal"
      }
    },
    "preview": {
      "distribution": "internal",
      "env": {
        "EXPO_PUBLIC_STOWAWAY_BUILD_AUDIENCE": "internal-personal"
      }
    },
    "production": {
      "distribution": "store",
      "autoIncrement": true,
      "env": {
        "EXPO_PUBLIC_STOWAWAY_BUILD_AUDIENCE": "public"
      }
    }
  }
}
```

This is a sketch of the build subsection; preserve all unrelated existing EAS settings. Static policy evidence may use repository-document references rather than embedding private permission correspondence in the app. Record exact acquisition scope, review date, review-due date and any actual authorization expiry separately. A review date is not an invented permission expiry.

For M0, only fixture/SB implementations are registered. Future providers can have policy catalog entries, but no callable placeholder importer. All SB initial import/resume/update/image actions require internal-personal. Public/unconfigured rejection occurs before delegation, request-client work, ownership or attempt-time mutation. Local reading, Read/locator writes, deletion and safe explicit external-source opening are separate from network acquisition permission.

The allowed path delegates once to the existing service with identical signal/progress/onSettled/options and result/error behavior. The facade never acquires another story owner. If a denied call promises onSettled, invoke and await it exactly once without an owner; the allowed call leaves it entirely to the underlying operation. Do not add an unconditional second settlement finally block. Preserve the screen's existing defensive final-refresh fallback, which may invoke the same refresh closure again after service

settlement; exactly-once here describes service/facade ownership, not an end-to-end screen callback count.

Guard every reachable acquisition route, including S1 diagnostics, and make the application import boundary explicit so screens cannot bypass the facade. Keep the underlying SB URL/manifest/image validators and shared pacing client intact. Internal experimentation never authorizes access-control or anti-bot bypasses.

Later public composition gate

M0 may still statically import SB implementation modules. It therefore **does not prove their exclusion from a public bundle**. Before any public build/TestFlight distribution, introduce a reproducible public composition containing only approved acquisition implementations, inspect the produced module graph/routes, and test the actual artifact. Separate the local image-file factory from modules that import the SB network client where necessary. Inspect Expo Router diagnostics and deep links too.

A future public registry can contain accepted EPUB/file operations while the internal registry additionally composes SB. There must be no remote flag, hidden switch or OTA permission unlock. A changed authorization/composition requires a reviewed release process. Checking only whether the word "SpaceBattles" appears in bundle text is insufficient: benign attribution may remain, while executable code may not contain that literal.

Keep production/public distribution non-shipping until this composition/evidence/content review passes. That release gate normally follows M2 and can accompany M3; it is not permission to expand M0 into submission work.

10. UI IMPACT

M0 preserves the current Library and Reader designs, chapter prominence, operation progress and SB labels. Replace source-specific eligibility checks with a pure derived action model and guarded dispatch. Preserve existing abort, stale-handler, focus-refresh, confirmation and ownership behavior.

When M1 is functional, change Add Story to **Add / Import** with two simple entry actions: **Paste URL** and **Import File**. Do not add a dead file button before the importer works. Unsupported/disabled URLs receive an actionable explanation without fetching them.

Add one concise source/origin line in Library or Contents, with expanded metadata only where useful. Generic EPUB says "EPUB" or "Imported EPUB". Unknown files say "Imported File" only if that type is actually supported. M3 may say "AO3 Download - imported file" only through an explicit recognized-export workflow with internally consistent source references. Filenames, one dc:source URL or a generator string alone are insufficient.

View Source appears only for a validated applicable HTTPS URL. Declared file source links are labeled as declared provenance, not authenticated origin. Keep rights/creator text and language as plain text. File deletion controls must not depend on sourceUrl being nonempty.

Action copy is truthful: SB keeps Check for Updates/Resume Download/Retry Images; EPUB uses Continue Import/Restore Reading Copy/Select Original File/Repair Images from Original as actually supported. A missing image does not suggest remote Retry Images for a local-only item. Counts may use "available offline" on mixed-origin surfaces; do not mechanically rename SB-specific accepted progress in M0.

The first EPUB profile discloses stripped formatting, unsupported resources and inactive internal links in a compact import result. Preserve the approved app styles, themes, 26-font catalog and partial font picker.

Setting a safe language on generated HTML is a small shared input, not a Reader redesign. No tab bar, discovery surface, Book style, pagination or source-specific renderer.

11. TEST PLAN

Use original synthetic EPUBs generated from small text/XML/assets. Do not commit full copyrighted books, private downloads, real captures or prose logs. Separate deterministic results, host SQL feasibility, release-bundle validation and user-reported physical evidence.

Layer	Required cases and assertion
M0 pure registry	Fixture/SB matrix; unknown origin; unsupported operation; explicit internal/public/missing/malformed audience; provenance cannot grant acquisition
M0 service composition	Allowed delegate called once with identical arguments/options/result/failure; denied delegate never called; no second ownership or facade-added settlement; preserve the screen fallback
M0 screens	Existing stale handler, busy/refresh, abort, native confirmation and settled refresh behavior remains; disabled acquisition does not hide local deletion
Source regression	Existing I0/I1/I2 pacing/refusal, complete manifest, duplicate resume, no implicit checks, exact source-image policy and canonical output tests remain
v6-to-v7 migration	All old rows/IDs/body bytes/hash/version/blocks/links/images/Read/locators/preferences/timestamps preserved; pending chapter mapping; rollback/retry; current-only uniqueness; S2A remains v1
Identity and ownership	Opaque local IDs; stored membership validated before and during queued local writes; stale story/chapter association rejected; ownership through final refresh
EPUB package	EPUB2/3, OPF/nav/NCX, front/back/auxiliary items, duplicate/missing titles, empty sections, repeated resource rejection, nested TOC, missing/unsupported primary resources
ZIP hostile fixtures	Limit and limit+1, forged sizes, high expansion, many entries, ZIP64/split/encrypted, descriptor variants, harmless explicit directory entries, overlaps, duplicate names, traversal, encoded aliases, truncated member and bad CRC
XML hostile fixtures	Malformed closing tokens, duplicate attributes, namespace collisions/unbound prefixes, deep nesting, many tiny nodes, oversized attributes, invalid UTF-8, UTF-16, internal entities/DTD/XInclude
Canonical content	No source script/CSS/remote resources; body-only extraction; fixture sentinel cannot be impersonated; stable output/hash/blocks; internal links inert; external links explicit HTTPS only
Images	Shared file, repeated instances, animation, MIME/magic mismatch, oversize/dimensions/pixels, missing optional resource, deterministic budget omissions, generated names, existing SB restrictions unchanged
Cancellation/storage	Cancel in copy/hash/stage/publish; force-quit boundaries; disk full; transaction failure; after-COMMIT error/readback; active-stage and referenced-file cleanup protections

Layer	Required cases and assertion
Recovery/deletion	Exact original restored; missing/stale/unsupported original; recipes survive Remove Reading Copy; full child deletion; shared assets survive other-story removal; cleanup warning is logical success
Read/Continue/reader	Pending/current/auxiliary state, body availability, retained locator shells, independent Read and locator saves, quiescence/Retry, font/theme/size restoration
Reimport later	Exact duplicate, explicit separate copy, ambiguous work match, stable resource rename/reorder, changed boundaries, rating/warning metadata, no silent overwrite or progress transfer
1,000+ chapters	Full body import, metadata-only lists, shared images, bounded memory, cancellation, reopen/Continue/last chapter, realistic long single chapter near cap
Public artifact later	Actual compiled/module composition, route/deep-link entry points, no disallowed acquisition, no remote unlock, local deletion/read unaffected
Standalone iPhone	Installed release JS; local Files selection/cancel/provider behavior, oversized selection and provider-copy failure; import and offline restart/read/images/Read/Continue; font controls; package recovery and deletion

M0 verification is deliberately bounded

Run the repository-required npm run check, npx expo install --check and npx expo-doctor, plus relevant iOS bundle/config export for internal and public audience values. Test denied dispatch before side effects and exact allowed-path callback ownership. Do not remove the screen's existing defensive final-refresh fallback to satisfy a simplistic total callback count. Reuse the existing integration suite rather than inventing dozens of tests that restate object literals.

Prepare a focused manual phone checklist: ordinary existing import/resume/update/image action, cancellation and settled UI, local removal confirmations, Continue, reader navigation and offline cold start. No full v0 acceptance restart is required merely because the facade exists. Record tests not exercised; do not manufacture live mutations or claim physical acceptance from an export.

M1a feasibility gate

Before normal-library EPUB persistence, verify pinned public imports and FileHandle/picker APIs in installed SDK57 typings, Metro and release Hermes. Run hostile XML and ZIP cases at all relevant limits on the actual target iPhone, record peak memory and longest uninterrupted processing task, and measure cancellation. The parser's synchronous allocation/stall risk is a blocking gate, not a footnote to fix after shipping.

The current 1,000-entry repository tests validate reconciliation/count/Read behavior; they do not prove 1,000 downloaded bodies or native importer performance. The global foreign_key_check per transaction may make total full-import work grow sharply. Preserve it initially, measure the real bottleneck and scope any optimization separately without weakening integrity/rollback. [R3, R11]

12. DEPENDENCY RECOMMENDATION

M0: add no dependencies and change no dependency versions.

The actual repository declares Expo ~57.0.27, React Native 0.86.3, React 19.2.3, expo-file-system ~57.0.7, expo-crypto ~57.0.3, expo-sqlite ~57.0.4, htmlparser2 ^12.0.0 and domhandler ^6.0.1. It has no ZIP dependency or strict package XML validator. These are manifest declarations; this review did not reinstall or re-resolve the lockfile. [R12]

Recommended M1a candidate stack

Pin exact versions in the feasibility slice. These are recommendations to investigate and gate, not claims of verified native compatibility.

Component	Proposed version/range	Maintenance/license	Compatibility, native and bundle implications
Raw DEFLATE	pako "3.0.2"	Current upstream release 12 Sep 2026; MIT AND Zlib; no runtime dependencies	Pure JS; public low-level buffer API. No added native module. Metro/Hermes public imports and real release delta must be measured
Incremental SHA-256	@noble/hashes "2.4.0"	Current upstream release 27 Aug 2026; MIT; no runtime dependencies	Import only public sha2.js SHA-256 path. ESM/export resolution must work. No native module; measure tree-shaken bundle result
Bounded XML document parse	@rgrove/parse-xml "5.0.0"	Release 30 Aug 2026; ISC; no runtime dependencies	ESM, browser-oriented XML1.0 well-formedness parser. No native module. One bounded tree, no streaming/early node quota; release Hermes gate required
File picker/read handles	Existing expo-file-system ~57.0.7	Existing Expo dependency; maintained Expo SDK57 API	Official docs include iOS File.pickFileAsync and seekable FileHandle. First choice; inspect installed overloads and test providers/cancellation/oversize copies before JS size checks
Picker fallback only	expo-document-picker, Expo SDK57-compatible ~57.0.3 resolved by npx expo install	Expo-maintained, MIT	Add only if installed FileSystem picker fails needed behavior. New native module requires a rebuilt standalone binary

Versions/licenses/API behavior come from upstream package metadata/releases and official Expo docs. [E7-E9, E11-E13] Do not substitute npm latest for Expo's compatible native-module resolution. Do not upgrade the application's SDK or add Node polyfills to make an unsuitable library fit.

Why these primitives?

ZIP: Pako 3 exposes ZStream and low-level zlibInflateInit2/zlibInflate/zlibInflateEnd. Caller-supplied output/avail_out permits a fixed-buffer raw-DEFLATE loop. Pako is not a ZIP container validator: Stowaway still needs a small audited ZIP32 header/index/range/descriptor/CRC32 implementation. Do not import private CRC code or collect one-shot inflate output. [E11]

Hash: Expo Crypto's documented digest APIs consume a complete BufferSource or string; they do not provide the incremental package-hash interface this design needs. Noble documents incremental SHA-256 updates. Continue using the accepted Expo digest path for existing small canonical content; use incremental hashing for archive streams without changing SB hashes. [E8, E12]

XML: htmlparser2 is deliberately forgiving; XML mode is not evidence of complete well-formedness validation. Keep it as part of the accepted sanitizer, after a separate package-validation boundary. parse-xml provides a bounded-document alternative, but has no documented early depth/node quota and does not load external DTDs or resolve their custom entities. M1a must provide and test namespace binding and expanded-name validation, including unbound/reserved prefixes and duplicate expanded attribute names; do not assume parse-xml supplies this behavior. [E13, E16]

Alternatives not selected: fflate's streaming ZIP API is attractive, but its inflater may resize output before delivering a consumer callback; counting bytes in ondata does not prove a pre-allocation bound. Worker-based options are not assumed to work in this RN runtime. saxes is archived; sax-js's current namespace/duplicate-attribute issues require caution. These are reasons for the selected feasibility investigation, not blanket claims that those packages are unsafe. [E14-E15]

Bundle/security decision: All three proposed core additions are inspectable pure-JS packages, but their exact Stowaway release-bundle and peak-memory costs are unmeasured. Record those costs in M1a. Do not quote browser gzip sizes as Metro binary measurements. Pin lockfile integrity, retain license notices, inspect the actual published public exports and run the hostile corpus. A custom ZIP reader is material security work; if that work or the synchronous XML parser cannot satisfy the gate, stop and select a safer primitive before proceeding.

13. PHASED ROADMAP

Milestone	Bounded result	Exit gate
M0: source action foundation	Pure capabilities/policy, explicit audience, guarded legacy dispatch and capability-aware existing screens	SB regression suite, denied-operation tests, internal/public exports; no schema/dependency/source addition
M1a: EPUB runtime/security proof	Pinned ZIP/hash/XML primitives, seekable I/O, synthetic package staging and device measurements	All resource/security/compatibility/cancellation gates; no normal-library migration
M1b: minimal persistence	v7 explicit entry IDs, two file sidecars, stored ownership, management and render cleanup compatibility	Preservation/rollback/S2A tests and native migration checks
M1c: first imported EPUB	New/duplicate/exact-package partial import, canonical spine text, bounded metadata and local static images	Same Library/Contents/Reader works across SB and file origins; no remote EPUB requests
M1d: lifecycle acceptance	Local repair, exact-original restore, truthful purge, deletion, offline cold restart	Original retained; body/read/locator/asset recovery proved; unsupported cases clearly reported
M2: backup/restore	Versioned local export/restore of durable multi-origin state and retained packages	Validated staging, checksums, collision/rollback and cold-start tests
M2 follow-up, separately scoped	Changed-package reimport and optional internal chapter/block links	Backup first; conservative identity matching and existing quiescence preserved
M3: AO3 official-download specialization	Explicit recognized EPUB/HTML download workflow and attribution	File provenance cannot enable live adapter; current source/content policy review
Public composition gate	Approved acquisition modules only, actual artifact inspection, release evidence	Required before any public or TestFlight distribution; no hidden unlock
M4: first approved direct adapter	One source whose permission and content requirements are met	Written/official scope evidence, source-specific access/assets contract and deterministic suite
M5+	Additional approved sources, one at a time	Reuse proven contracts without loosening global policy

The user-proposed sequence is sound. EPUB should precede full backup implementation because the actual retained package/recipe model informs a useful backup format. Backup should follow immediately and precede destructive changed-file reimport, broader testers or public distribution.

Backup/restore scope

Use a versioned manifest plus consistent database/canonical export, retained EPUB packages, validated images, Read flags, locators, preferences, import recipe versions, identities and checksums. Exclude disposable Reader renders, transient ownership and unfinished staging unless explicitly represented as incomplete.

Quiesce operations and obtain a supported consistent SQLite snapshot/export; do not copy an open database while ignoring its WAL. Validate the entire backup and supported schema/import versions into staging before replacing live state. Restore to a new temporary database/file namespace, verify

relationships/checksums, then swap through an explicit recoverable commit process. Preserve the old state until success is confirmed.

Restored source descriptors/provenance may inform display, but **current compiled policy determines capabilities**. A backup cannot restore an old public permission, active operation, source cookie or background task. Missing optional images can be reported; missing required canonical state or a corrupt manifest cannot be accepted as a complete restore. No cloud sync, accounts, server or collaborative state is needed.

14. RISKS / DEFERRED ITEMS

Actual requirements for the next source

- A real capability/service boundary with fail-closed acquisition policy.
- Explicit persistent chapter identity and stored-owner validation before non-forum IDs enter the normal library.
- Minimal non-destructive schema evolution, including management version/snapshot/deletion changes.
- Bounded archive/XML processing and complete validated staging before publication.
- An immutable retained original and truthful exact-package recovery.
- Source-free canonical rendering, static-image validation, generated paths and reference-aware cleanup.
- Deterministic synthetic fixtures plus standalone native acceptance.
- Backup immediately after EPUB and current-source/public-artifact review before distribution.

Material remaining uncertainty

The proposed package stack has not run in this repository's Hermes/iPhone binary. XML input caps do not automatically solve synchronous tree allocation; ZIP callbacks do not automatically bound inflater memory. The feasibility gate may require a different parser or archive primitive. That uncertainty does not block M0.

The existing global FK check may become expensive with thousands of body transactions and new recipe rows. Do not optimize speculatively or remove the check. Measure after the first valid full-body importer exists.

A constrained prose importer will reject or simplify some legal EPUBs: ZIP64, UTF-16, fixed-layout dependence, unsupported primary media, uncommon entities, repeated spine resources, complex CSS and navigation-only content. The initial internal-link limitation may be unacceptable for footnote-heavy books. Make these limitations visible and add capabilities only through bounded subsequent work.

Retained packages increase disk use and can still be lost through user deletion of the app or storage failure. They are not a backup. Image byte/dimension limits are structural safeguards, not a proof of decoder correctness.

Source permissions and App Store content obligations remain separate from engineering completeness. The present baseline establishes no new written permission. AO3 Mature handling needs correction in any future explicit-content exclusion rule.

Deliberately deferred

No aesthetic schema renaming, universal adapter/plugin marketplace, generic browser/scraper, ORM or global state framework. No account/backend/cloud sync, recommendations/discovery/social/comments, annotations/highlights/TTS, pagination/Book style, authenticated/private-source scraping, automatic refresh, DRM circumvention or App Store submission UI.

Do not infer comprehensive accessibility, natural live mutation, storage-failure or transaction-boundary phone coverage from accepted v0. Keep the repository's exact physical-evidence boundaries. The current accepted source/reader product is not reopened merely because future abstractions could be cleaner.

15. EXACT FIRST IMPLEMENTATION SLICE

M0 - capability and acquisition-policy boundary for the accepted app

Purpose: Make the next ingestion path possible without changing existing storage or reading behavior. Centralize source eligibility and service dispatch, prove current SB behavior remains exact, and prevent public/unconfigured acquisition.

In scope:

1. Add pure BuildAudience, origin projection, source descriptor, capability/action and policy-decision types.
2. Register only implemented fixture and SpaceBattles behavior. Keep future source policy entries non-callable, or omit them until needed.
3. Add an explicit build audience and committed EAS profile values; preserve current internal development commands through a tiny cross-platform launcher.
4. Compose a guarded native operation facade around existing import/resume, update, image and management services. Keep local management independent of acquisition permission.
5. Replace source-specific action-eligibility predicates and direct native acquisition imports in Add/Contents with facade/capability calls. Preserve existing UI layout and labels.
6. Guard S1 diagnostic acquisition by explicit internal audience as well as its current development guard. Preserve S2A isolation and behavior.
7. Add focused policy/dispatch/callback/race regression tests and document M0's limits.

Likely files: New src/domain/source-capabilities.ts, src/sources/registry.ts, src/sources/operations.ts (pure/testable dispatch factory), src/sources/native-operations.ts (native composition), src/config/build-audience.ts, and a small scripts/expo-internal.mjs if needed. Narrow edits to app/add-story.tsx, app/story.tsx, source diagnostic entry, relevant operation/failure presentation, eas.json, package.json scripts and existing lint/import-boundary configuration. Add focused tests and one M0 document; update README/AGENTS only to the actual accepted scope.

Names are suggestions, not authorization to reorganize directories. Existing importing/native.ts and images/native.ts can remain internal implementation modules. Make the app boundary reviewable so no route bypasses policy; do not create a second request client or move request pacing.

Schema impact: None. user_version remains 6. No migration, backfill or repository writer refactor.

Dependency impact: None. No installs/upgrades/lockfile churn. package.json may change scripts only.

Reader/content impact: None. No Reader, bridge, locator, sanitizer, identity constructor, image URL rule or canonical payload changes.

Acceptance criteria

- In explicit internal-personal mode, all accepted SB operations delegate exactly once with existing options and behavior.
- Fixture has no network/removal capability. Unknown origins and unsupported operations fail closed.
- Public/unconfigured/invalid audiences cannot call SB import, resume, update, source-image discovery/retry or S1 network work.
- Denial occurs before native delegate work, ownership or update-attempt mutation. The denied facade's promised settlement occurs once; allowed-service settlement and the screen's defensive refresh fallback remain unchanged.
- Existing local reading, Read/locator writes, Continue and confirmed local removal are not disabled by network policy.
- Screens retain stale-confirmation, busy/focus, cancellation and authoritative-refresh behavior.
- The existing deterministic suite and required project checks pass without weakening assertions. Relevant internal/public audience exports succeed.
- M0 is documented as an internal foundation with service gating, not proof of public source-module exclusion or public-release readiness.

Device checks and stop rules

Prepare a focused standalone/internal smoke checklist for existing actions and Metro-free offline reading. If no phone is available, record that limitation; do not turn export success into physical acceptance. No new paid EAS build or public distribution is part of the coding slice unless separately authorized.

Stop and report if the implementation requires schema/content/reader changes, dependency upgrades, source permission changes, a new source, a new request client or ownership relocation. A failing preserved invariant is a defect to resolve within M0 or a reason to narrow it, not permission to weaken the invariant.

Complete one implementing session, then a separate sequential diff/security review. The review must inspect the final diff and ownership/callback paths, not run concurrently with edits. Stop after the scoped patch, evidence and review result. No automatic commit, push, PR or merge is included in this proposed handoff.

16. EXACT FIRST CODING TASK

Create and test the pure source-capability and policy model for the existing fixture and SpaceBattles origins.

Add src/domain/source-capabilities.ts and src/sources/registry.ts, plus focused tests. Project current StoryMetadata into a discriminated fixture/remote origin without changing its persisted shape. Define explicit BuildAudience and a decision result that distinguishes allowed, policy-disabled and unsupported-operation/source.

Assert the full fixture/SB action matrix for internal-personal, public, missing and malformed audiences. Test that local reading/state/deletion eligibility is independent of network acquisition policy, and that

declared file provenance cannot grant capabilities. Use a test-only file-shaped value if useful; do not register an EPUB implementation.

The first patch must contain no network/file/database initialization, reader/sanitizer edits, migration, dependency, source fetch or UI change. Once this pure model is verified, continue only with the remaining M0 facade/config/screen steps in section 15. Do not begin M1.

17. IMPLEMENTATION HANDOFF PROMPT

Use the following prompt in a subsequent implementing session. It authorizes only the local M0 patch and verification when submitted for that purpose. The present report-producing session has made no repository changes.

STOWAWAY M0 IMPLEMENTATION HANDOFF

Capability and acquisition-policy boundary Architecture baseline: 10 October 2026

This is a ready-to-run prompt for a subsequent implementing session. The architecture/research session that produced it was read-only.

Implement only M0 for repository owencchapman24/stowaway.

BASELINE

The reviewed main commit is: d43cf0963497e40836cac40f0c2ed2df43a0262d

Verify current main and the working tree before editing. If main advanced, read the diff and adapt only when it does not change this slice's assumptions. Do not reset, overwrite or discard unrelated user work. Use an isolated local implementation checkout/worktree when needed. This prompt authorizes the local M0 patch and its verification. Do not commit, push, create a PR, merge, start a paid EAS build, alter signing or distribute a public/TestFlight build.

Stowaway's SpaceBattles-only personal v0 is accepted. Preserve it. This is not an EPUB implementation or another general v0 hardening pass.

READ FIRST

Read AGENTS.md and docs/Stowaway_October_2026_Groundwork_Audit.md first. Read all of the following completely before implementation:

- README.md
- docs/Pre_UI_Hardening.md
- docs/Pre_S2B_Hardening.md
- docs/S2B_Standalone_Acceptance.md
- docs/UI_UX_Implementation.md
- docs/Stowaway_Public_Release_Readiness_and_Source_Permissions.md
- docs/I0_Real_Story_Import.md
- docs/I1_Manual_Updates.md
- docs/I2_Source_Images.md
- docs/I3_Read_State.md
- docs/I4_Library_Management.md

Read the accompanying Stowaway Multi-Source Architecture Report if available. This prompt remains sufficient for M0 if that report is not attached.

Trace actual production code, especially:

- src/domain/types.ts and identity.ts
- src/importing/import-story.ts, native.ts, ownership.ts, failure-presentation.ts
- src/images/native.ts, download.ts and source-url.ts
- src/sources/spacebattles/native.ts, request.ts and url.ts
- src/storage/repository.ts, management.ts, management-native.ts, availability.ts and migrations.ts
- src/app/add-story.tsx, story.tsx, index.tsx and s1.tsx
- src/ui/operation-copy.ts
- existing ownership, operation, I0-I4 and UI integration tests
- package.json, eas.json and existing lint configuration

PURPOSE

Add a narrow source-neutral action boundary around existing behavior:

1. Pure origin/capability/policy decisions.
2. Explicit internal/public build audience.
3. Guarded dispatch to existing services.
4. Capability-based screen eligibility with all existing race controls intact.

Do not require local files to implement network methods. Do not register a pretend EPUB importer. Future file provenance never implies a live adapter.

EXACT FIRST CODING TASK

Create src/domain/source-capabilities.ts and src/sources/registry.ts, with focused tests, or equivalently small files consistent with the repo.

Define:

- BuildAudience = internal-personal | public | unconfigured.
- A discriminated fixture/remote origin projection from current StoryMetadata.
- Typed supported operations/action descriptions.
- Explicit policy statuses: internal-only, public-approved, public-file-import-only and disabled.
- A pure eligibility result distinguishing allowed, policy-disabled and unsupported-source/operation.

Only fixture and SpaceBattles have implemented registry entries. A policy catalog may describe future sources, but it cannot expose callable operations.

Test fixture/SB capabilities under internal-personal, public, absent and invalid audience values. Test unknown origins, unsupported operations, local-management independence and rejection of capability escalation from declared provenance. A test-only file-shaped input is permitted; no production file-import stub.

This first patch must have no network, file or database initialization, UI change, migration, dependency, identity-constructor change or reader/sanitizer change. Verify it, then complete the remaining M0 work below. Do not start M1.

REMAINING M0 SCOPE

A. BUILD AUDIENCE

Use a strict, statically embedded public environment value:

EXPO_PUBLIC_STOWAWAY_BUILD_AUDIENCE

Allowed values:

- internal-personal
- public

Missing, empty or malformed values resolve to unconfigured and deny acquisition. Do not infer audience from __DEV__, NODE_ENV, a profile name at runtime, package metadata or a user setting. Internal preview also runs release JavaScript.

Add explicit values in eas.json:

- development: distribution internal, audience internal-personal
- preview: distribution internal, audience internal-personal
- production: distribution store, audience public

Preserve existing CLI version, app-version settings, autoIncrement and all unrelated configuration. No signing, bundle-ID or native-permission changes.

Use a direct static process.env.EXPO_PUBLIC_STOWAWAY_BUILD_AUDIENCE reference so Expo can inline it. Keep pure parsing testable separately.

Preserve ordinary internal development commands with a small cross-platform Node launcher if needed. Avoid shell-only environment syntax that breaks Owen's Windows workflow; do not add a package merely to assign an environment value. package.json may change scripts only. No dependency or lockfile changes.

B. GUARDED OPERATIONS

Prefer a small testable dispatch factory in src/sources/operations.ts and a native composition module in src/sources/native-operations.ts.

Wrap existing:

- importNativeStory, including its resume/redownload path
- updateNativeStory
- downloadNativeImages, including discovery/retry
- native management dispatch where useful for capability-based screen use

Keep native service modules as internal implementation details. Make the app's import boundary reviewable so routes cannot bypass policy. Guard S1 native diagnostic acquisition by explicit internal audience as well as its existing development-only guard. S2A remains isolated and unchanged.

For allowed SB acquisition:

- Delegate exactly once.
- Forward signal, onProgress, onSettled and existing options unchanged.
- Preserve existing results and errors; do not catch/relabel allowed failures.
- Do not acquireStory around a delegate that already owns the story.

- Do not create another RequestClient, queue or pacing wrapper.
- Do not move final authoritative refresh or ownership release.

For denied acquisition:

- Reject before calling any native delegate.
- No source request, image allocation, story ownership or attempt timestamp.
- Return a typed policy-disabled/unsupported result with useful presentation.
- If the facade contract promises onSettled, await it exactly once on denial. On the allowed path, the existing delegate owns that callback entirely.
- Do not add an unconditional wrapper finally that calls onSettled again.
- Preserve the screen's existing defensive final-refresh fallback. Do not treat a total end-to-end callback count as a reason to remove accepted screen logic.

Capabilities do not replace SB URL, identity, category, COMPLETE manifest, same-origin image, redirect or access-refusal validation.

Do not gate local reading, Read/Unread, locators, Continue, deletion or cleanup on remote acquisition permission. An existing SB item in a public/unconfigured context must remain locally readable and removable. Safe explicit opening of a validated source URL is separate from adapter acquisition.

C. SCREEN ADAPTATION

Replace source-specific eligibility checks and direct native acquisition imports in Add Story and Contents with the facade/action model.

Preserve:

- current layouts and SpaceBattles labels
- existing compound busy/active/focus/refresh predicates
- stale-handler and retained-confirmation guards
- native confirmations and their existing count
- abort behavior and no early UI release
- authoritative onSettled refresh before release
- no operation on screen open
- current navigation and local read-state handling

A static capability bit must not replace race/ownership checks. Service dispatch must check eligibility again. Do not add Add / Import or a disabled file button yet; that UI belongs to a functioning M1 importer.

D. DOCUMENTATION

Add one concise M0 implementation/evidence note. Update current status pointers only as warranted by actual results. State: "M0 adds fail-closed acquisition eligibility and preserves the existing internal SpaceBattles workflow. It does not establish public source-module exclusion or public-release readiness."

Future source approval needs separate evidence and a reviewed public build composition. A boolean gate or hidden button is not proof of module exclusion. No hidden remote flag or OTA permission unlock.

PRESERVE ALL ACCEPTED INVARIANTS

Do not change:

- existing story/chapter IDs or numeric legacy constructors

- schema v6, migrations, repository writer algorithms or fixture seed behavior
- canonical sanitized HTML, hash, sanitizer version, generated blocks or links
- source-image host restrictions, validation, staging or cleanup algorithms
- Read flags, availability, saved-unlisted behavior, Continue or locator ordering
- same-story operation ownership and authoritative refresh before release
- route-owned ReadCompletion
- prepareForExit/quiesced navigation, acknowledgement and timeout behavior
- fonts, themes, typography, reader template, bridge or style design
- S2A's exact dedicated resources and v1 behavior
- shared SB client pacing, refusal handling or explicit-only operations

No accounts/cloud/backend/discovery, generic scraper engine, plugin framework, ORM/global state library, public adapter, EPUB/HTML parser, backup implementation, automatic refresh, authenticated scraping or DRM work.

TESTS AND VERIFICATION

Add meaningful focused tests for:

1. Internal/public/unconfigured registry matrix.
2. Unknown origin and unsupported operation; provenance cannot escalate access.
3. Allowed delegate once with unchanged options/results/errors.
4. Denied delegate never invoked and no operation side effects.
5. The allowed service retains sole ownership of its settlement callback; the facade adds none. A denied facade invokes its promised callback once. Preserve the screen's defensive final-refresh fallback, even when it invokes the same refresh closure again. Ownership remains through service settlement.
6. Screen eligibility retains existing busy/stale/confirmation constraints.
7. Local reading/state/removal remain independent of acquisition policy.
8. Explicit development/preview/production audience configuration.
9. S1 cannot bypass the audience guard.

Run the repository's required:

- npm run check
- npx expo install --check
- npx expo-doctor
- relevant iOS exports for explicit internal and public audience values

Use the repository's supported Node version. On Windows use npm.cmd/npx.cmd where PowerShell script policy requires it. Keep generated exports out of git. Use synthetic fixtures only; no live story requests in tests, CI or the agent's implementation/review pass. Do not weaken tests to achieve green results.

Prepare a focused manual standalone/internal phone checklist:

- existing import/resume/update/image actions and cancellation settlement
- existing local removal confirmation and cleanup presentation
- Continue, Previous/Next, Read/Unread and font/theme controls

- Metro-free airplane-mode cold start and saved local content/state

Record phone checks as pending unless actually performed and reported. Do not turn a bundle export into physical acceptance or restart the entire accepted v0 acceptance program. No paid EAS build is authorized by this prompt.

REVIEW AND STOP

After edits and verification, perform a separate sequential diff/security review. Do not review a moving diff concurrently with implementation. Check all reachable acquisition routes, callback ownership, stale UI guards and accidental changes to protected files/invariants.

Stop and report if M0 would require a migration, Reader/sanitizer change, dependency upgrade, new source, ownership relocation or source-permission change. Resolve defects within scope; do not silently expand it.

Return:

- concise explanation of the patch and resulting behavior
- exact files changed
- verification commands/results and evidence limits
- final sequential review findings
- remaining manual phone checks
- confirmation that schema/dependencies/content/reader invariants were preserved

Stop after the M0 patch and review. Do not begin EPUB, commit, push, create a PR, merge, build/distribute publicly or request source permission.

Evidence register

Repository references below are the actual implementation/document locations at the reviewed SHA, not citations to the historical project attachments. All GitHub reads were read-only. Common pinned base:

https://github.com/owencchapman24/stowaway/tree/d43cf0963497e40836cac40f0c2ed2df43a0262d

[R1] Accepted state and evidence boundaries: AGENTS.md; README.md.

[R2] Domain: src/domain/types.ts; src/domain/identity.ts.

[R3] Storage and migration: src/storage/repository.ts; migrations.ts; database.ts; availability.ts.

[R4] Removal and files: src/storage/management.ts; management-native.ts; local-files.ts; render-cleanup.ts; file-primitives.ts; files.ts.

[R5] Import/ownership: src/importing/import-story.ts; native.ts; ownership.ts; failure-presentation.ts.

[R6] SpaceBattles adapter: src/sources/spacebattles/url.ts; parse.ts; request.ts; diagnostic.ts; native.ts; types.ts.

[R7] Canonical content: src/content/sanitize.ts.

[R8] Images: src/images/source-url.ts; download.ts; files.ts; validate.ts; slots.ts; repository.ts; render.ts; native.ts.

[R9] Reader: src/app/reader.tsx; src/reader/ChapterReader.tsx; template.ts; bridge.ts; read-state.ts; position.ts; navigation.ts; appearance.ts.

[R10] Screens and copy: src/app/add-story.tsx; story.tsx; index.tsx; s1.tsx; s2a.tsx; src/ui/operation-copy.ts and shared UI components.

[R11] Tests: tests/integration/i0-import.test.ts; i1-update.test.ts; i2-images.test.ts; i3-read-state.test.ts; i4-management.test.ts; storage.test.ts; s2a-recovery.test.ts, plus reader/ownership/UI tests and synthetic fixtures.

[R12] Runtime/build: package.json; eas.json; app.json; existing lint/TypeScript/Jest configuration.

[R13] All requested engineering documents read: docs/Stowaway_October_2026_Groundwork_Audit.md; Pre_UI_Hardening.md; Pre_S2B_Hardening.md; S2B_Standalone_Acceptance.md; UI_UX_Implementation.md; I0_Real_Story_Import.md; I1_Manual_Updates.md; I2_Source_Images.md; I3_Read_State.md; I4_Library_Management.md.

[R14] Permission baseline: docs/Stowaway_Public_Release_Readiness_and_Source_Permissions.md; related bounded source context in docs/S1_Source_Policy.md.

[R15] Current successful Stowaway checks run for the reviewed main:

https://github.com/owencchapman24/stowaway/actions/runs/38022524764

Official external sources, accessed 10 October 2026:

[E1] W3C EPUB 3.3, Recommendation 13 January 2026.

https://www.w3.org/TR/epub-33/

[E2] W3C EPUB Reading Systems 3.3, conformance and supported package behavior.

https://www.w3.org/TR/epub-rs-33/

[E3] Official legacy OPF 2.0.1, packaging, metadata, manifest, spine and NCX. The historical filename contains "_draft" although the displayed document is a Recommended Specification.

https://idpf.org/epub/20/spec/OPF_2.0.1_draft.htm

[E4] Official legacy OPS 2.0.1, content documents, XML, links, CSS and EPUB2 features.

https://idpf.org/epub/20/spec/OPS_2.0.1_draft.htm

[E5] W3C EPUB 3.4, Candidate Recommendation Draft 2 October 2026.

https://www.w3.org/TR/epub-34/

[E6] PKWARE ZIP APPNOTE, archive fields/structures, methods, descriptors and integrity.

https://pkware.cachefly.net/webdocs/casestudies/APPNOTE.TXT

[E7] Expo FileSystem API, current SDK57 package version, picker and FileHandle.

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End of read-only report. Recommendations and code/SQL sketches are proposals; they have not been applied to the repository.